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MOLTEN SALT HEAT EXCHANGER

Abstract

A molten salt reactor system includes a fuel salt system and a coolant salt system. The fuel salt system is configured to circulate a molten fuel salt through a molten salt loop. The coolant salt system is configured to circulate a coolant salt through a coolant loop. The molten salt reactor system further includes a heat exchanger arranged along the molten salt loop and the coolant loop. The heat exchanger is configured to transfer heat from the molten fuel salt of the molten salt loop to the coolant salt of the coolant loop. The heat exchanger is configured to permit gravitational draining of at least one of the molten fuel salt or the coolant salt upon a shutdown event.

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Background/Summary

RELATED APPLICATION [0001] The present application relates and claims priority to U.S. Provisional Application No. 63/561,853, filed on Mar. 6, 2024, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The described examples relate generally to systems, devices, and techniques for heat exchangers used in a molten salt system.

BACKGROUND

[0003] Heat exchangers of various types are used in molten salt systems. A molten salt system may include a molten salt nuclear reactor system whereby a molten “fuel” salt is heated through nuclear fission reactions. One or more heat exchangers may be deployed in the molten salt system in order to transfer heat from the fuel salt to another medium, such as a coolant salt and/or other coolant medium. The molten fuel salt may include radioactive material, as may the coolant salt and/or other coolant medium. Such molten salt systems may operate continuously for prolonged periods, such as a period of years. Notwithstanding, the one or more heat exchangers deployed in the molten salt system may require maintenance, repair or replacement from time to time, and therefore, the molten salt system may be taken offline to facilitate such activity. Upon taking the molten salt system offline, however, the molten fuel salt and the coolant salt may begin to rapidly cool and solidify. Solidification of the molten fuel salt, coolant salt, and/or other materials in the heat exchanger could hinder the ability to maintain, repair, or replace said heat exchanger, particularly where the solidified materials include radioactive material. As such, there is a need for systems and techniques to facilitate the maintenance, repair, and replacement of heat exchangers in molten salt systems.

SUMMARY

[0004] In one example, a molten salt reactor system is disclosed. The molten salt reactor system includes a fuel salt system configured to circulate a molten fuel salt through a molten salt loop. The molten salt system further includes a coolant salt system configured to circulate a coolant salt through a coolant loop. The molten salt system further includes a heat exchanger arranged along the molten salt loop and the coolant loop. The heat exchanger is configured to transfer heat from the molten fuel salt of the molten salt loop to the coolant salt of the coolant loop. The heat exchanger is configured to permit gravitational draining of at least one of the molten fuel salt or the coolant salt upon a shutdown event.

[0005] In another example, the heat exchanger may include an outer shell defining a shell volume configured to receive one of the molten fuel salt or the coolant salt. The heat exchanger may further include an internal tube positioned at least partially within the shell volume and configured to receive the other of the molten fuel salt or the coolant salt.

[0006] In another example, the outer shell may include a baffle defining a shell circulation path about a portion of the internal tube positioned in the shell volume. In this regard, the heat exchanger may be arranged on the molten salt loop and the coolant loop with the circulation path positioned off-set from a horizontal direction of the molten salt reactor system.

[0007] In another example, the internal tube may extend oblique to the outer shell within the shell volume.

[0008] In another example, the internal tube may include a plurality of bends within the internal tube volume. In this regard, the oblique orientation of the internal tube relative to the outer shell may permit draining of the other of the molten fuel salt or the coolant salt upon the shutdown event.

[0009] In another example, the internal tube may extend substantially along the shell circulation

path.

[0010] In another example, the shell circulation path may proceed along a U-shaped path through the shell volume. The internal tube may include a U-shaped tube extending along the U-shaped path. Further, an inlet and an outlet for the shell circulation path and an inlet and outlet for the internal tube may each be arranged along a bottom side of the outer shell such that the respective inlets and outlets are each substantially aligned with a vertical direction of the molten salt reactor system.

[0011] In another example, the shutdown event may be triggered based on one or more of a loss of power, a high radiation alarm and/or a containment breach.

[0012] In another example, the molten salt reactor system may include a reactor vessel, a pump, and a drain tank all arranged along the molten salt loop. Further, the reactor vessel may output an elevated temperature form of the molten salt to the heat exchanger.

[0013] In another example, the heat exchanger may include a pair of oblique tubes associated with one another inside of an outer shell. A first oblique tube of the pair of oblique tubes may be configured to receive one of the molten fuel salt or the coolant salt. A second oblique tube of the pair of oblique tubes may be configured to receive the other of the molten fuel salt or the coolant salt.

[0014] In another example, the first oblique tube may be removably couplable from the outer shell and separatable from the first oblique tube.

[0015] In yet another example, the molten salt reactor system further includes a flange assembly operable to facilitate removal of the heat exchanger from the molten salt reactor system.

[0016] In another example, a molten salt reactor system is disclosed. The system includes a fuel salt system configured to circulate a molten fuel salt through a molten salt loop. The system further includes a coolant salt system configured to circulate a coolant salt through a coolant loop. The system further includes a heat exchanger arranged along the molten salt loop and the coolant loop and configured to transfer heat from the molten fuel salt of the molten salt loop to the coolant salt of the coolant loop. The heat exchanger includes at least one oblique tube configured to route at least one of the molten fuel salt or the coolant salt out of the heat exchanger upon a shutdown event.

[0017] In another example, the heat exchanger may include a shell and tube heat exchanger. Further, the at least one oblique tube may be arranged oblique to an outer shell of the shell and tube heat exchanger.

[0018] In another example, the molten fuel salt and the coolant salt may begin to solidify at a solidification temperature that is below 550° C.

[0019] In another example, the at least one oblique tube may include a first oblique tube configured to route the molten fuel salt out of the heat exchanger, and a second oblique tube configured to route the coolant salt out of the heat exchanger.

[0020] In another example, a portion of the first oblique tube and a corresponding portion of the second oblique tube may be removably interposed with one another with an outer shell of the heat exchanger.

[0021] In yet another example, the molten salt reactor system further includes a flange assembly operable to facilitate removal of the heat exchanger from the molten salt reactor system.

[0022] In another example, a method of draining a heat exchanger of a molten salt reactor system is disclosed. The method includes operating a fuel salt system by circulating a molten fuel salt through a molten salt loop. The method further includes operating a coolant salt system by circulating a coolant salt through a coolant salt loop. The method further includes transferring heat, using a heat exchanger, from the molten fuel salt of the molten salt loop to the coolant salt of the coolant salt loop. The method further includes gravitationally draining, upon a shutdown event, the molten fuel salt and the coolant salt from the heat exchanger.

[0023] In another example, the heat exchanger may include at least one oblique tube configured to

route at least one of the molten fuel salt or the coolant salt through the heat exchanger during the transferring.

[0024] In another example, said at least oblique tube may be arranged oblique to an outer shell of the heat exchanger.

[0025] In another example, the molten fuel salt may include a fissile material.

[0026] In addition to the example aspects described above, further aspects and examples will become apparent by reference to the drawings and by study of the following description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] FIG. 1 depicts an example molten salt reactor system.

[0028] FIG. 2A depicts an example heat exchanger of the molten salt reactor system of FIG. 1.

[0029] FIG. 2B depicts a cross-sectional view of the heat exchanger of FIG. 2A, taken along line 2B-2B of FIG. 2A.

[0030] FIG. 3A depicts another example heat exchanger.

[0031] FIG. 3B depicts a cross-sectional view of the heat exchanger of FIG. 3A, taken along lines 3B-3B of FIG. 3A.

[0032] FIG. 4A depicts another example heat exchanger.

[0033] FIG. 4B depicts a cutaway view of the heat exchanger of FIG. 4A.

[0034] FIG. 5A depicts another example heat exchanger.

[0035] FIG. 5B depicts a cross-sectional view of the heat exchanger of FIG. 5A, take along line 5B-5B of FIG. 5A.

[0036] FIG. 5C depicts an exploded view of the heat exchanger of FIG. 5A.

[0037] FIG. 6A depicts an example installation of an example heat exchanger in a molten salt reactor system.

[0038] FIG. 6B depicts another view of the example installation of FIG. 6A.

[0039] FIG. 7 depicts a flow diagram of a method of draining a heat exchanger of a molten salt reactor system.

[0040] The use of cross-hatching or shading in the accompanying figures is generally provided to clarify the boundaries between adjacent elements and also to facilitate legibility of the figures.

Accordingly, neither the presence nor the absence of cross-hatching or shading conveys or indicates any preference or requirement for particular materials, material properties, element proportions, element dimensions, commonalities of similarly illustrated elements, or any other characteristic, attribute, or property for any element illustrated in the accompanying figures.

[0041] Additionally, it should be understood that the proportions and dimensions (either relative or absolute) of the various features and elements (and collections and groupings thereof) and the boundaries, separations, and positional relationships presented therebetween, are provided in the accompanying figures merely to facilitate an understanding of the various embodiments described herein and, accordingly, may not necessarily be presented or illustrated to scale, and are not intended to indicate any preference or requirement for an illustrated embodiment to the exclusion of embodiments described with reference thereto.

DETAILED DESCRIPTION

[0042] The description that follows includes sample systems, methods, and apparatuses that embody various elements of the present disclosure. However, it should be understood that the described disclosure may be practiced in a variety of forms in addition to those described herein.

[0043] The following disclosure relates generally to heat exchangers for a molten salt system. A molten salt system may include a molten salt nuclear reactor system whereby a molten “fuel” salt is heated through nuclear fission reactions. One or more heat exchangers may be deployed in the

molten salt system in order to transfer heat from the fuel salt to another medium, such as a coolant salt or other coolant medium. The molten fuel salt may include radioactive material, as may the coolant salt or other coolant medium. Such molten salt systems may operate continuously for prolonged periods, such as a period of years. Notwithstanding, the one or more heat exchangers deployed in the molten salt system may require maintenance, repair or replacement from time to time, and therefore, the molten salt system may be taken offline to facilitate such activity. Upon taking the molten salt system offline, however, the molten fuel salt and the coolant salt may begin to rapidly cool and solidify. Solidification of the molten fuel salt, coolant salt, and/or other materials in the heat exchanger could hinder the ability to maintain, repair, or replace said heat exchanger, particularly where the solidified materials include radioactive material. Conventional heat exchangers may implement heat exchangers whereby the molten salt materials and/or other materials are susceptible to solidifying in the heat exchanger or otherwise to causing the heat exchanger to be inoperable and/or irreparable on shutdown of the molten salt system.

[0044] To mitigate these and other challenges, a molten salt system is disclosed herein with a heat exchanger that permits draining of materials (e.g., fuel salt, coolant salt, and/or other materials) on shutdown (e.g., a shutdown event) of the molten salt system. As used herein, a “shutdown event” may include any event in which the molten salt system ceases to circulate molten salt and/or otherwise ceases to produce nuclear reactions therefrom or therewith, including planned and unplanned shutdowns of the molten salt system. For example, the heat exchangers disclosed herein may include at least one oblique tube that is configured to route at least one of a molten fuel salt or a coolant salt out of the heat exchanger upon a shut down event of the molten salt system. The oblique tube may be oblique to a horizontal axis of the heat exchanger, which may thereby encourage any materials held therein to drain from the heat exchanger upon such shutdown event. As the fuel salt, coolant salt and/or other materials may be readily drained from the heat exchanger, the heat exchanger may be clear or substantially clear of (or have a reduced amount of) radioactive material therein upon the shutdown event. Accordingly, the reduced level of radioactive material in the heat exchanger upon the shutdown event may support the maintenance, repair, and replacement of the heat exchanger, in part, by reducing the risk of handling the heat exchanger due to said reduced level of radioactive material therein. Additionally, the heat exchanger may be installed within the molten salt system such that it can be removed without requiring substantial dismantling of said system.

[0045] To facilitate the foregoing, the heat exchangers of the present disclosure may be a shell and tube type heat exchanger. The shell and tube type heat exchanger may, in one example, include an outer shell and an inner tube arranged within the shell. The outer shell may be configured to circulate a first medium therethrough, such as a fuel salt. The inner tube may be configured to circulate second medium therethrough, such as a coolant salt. The inner tube may be positioned within the outer shell such that the fuel salt (or other medium) bathes the inner tube with the fuel salt, and thereby promotes the transfer of heat from the fuel salt to the coolant salt held in the inner tube. In one example, the shell and tube type heat exchanger may be configured to permit gravitational draining of the fuel salt and coolant salt therein. For example, the inlet and outlet of the outer shell may be arranged oblique to or otherwise offset from a horizontal axis of the heat exchanger. In this regard, upon the occurrence of a shutdown event, fluid (e.g., a molten fuel salt) held therein may be encouraged to drain from the outer shell, as opposed to otherwise solidifying within the outer shell. As a further example, the inlet and outlet of the inner tube may additionally or alternatively be arranged offset from a horizontal axis of the heat exchanger. In this regard, upon the occurrence of a shutdown event, fluid (e.g., a molten fuel salt) held therein may be encouraged to drain from the inner tube, as opposed to otherwise solidifying within the inner tube. In some cases, the respective inlet and outlet of the outer shell and inner tube may be arranged along a common direction to facilitate draining of the fluids contained therein along said common direction. Further, in some cases, one or both of the respective inlets and outlets of the shell and

tube may be arranged substantially along a vertical axis of the heat exchanger, which may facilitate draining of fluids (molten salts) contained therein along said vertical axis upon a shutdown event. [0046] In one example, the heat exchangers of the present disclosure may be installed into a molten salt system such that they can be removed for maintenance following drainage of the coolant medium therein. In this regard, the heat exchanger can be removed without requiring substantial dismantling of the molten salt system. For example, the heat exchanger may be installed via a flange assembly penetrating the reactor enclosure.

[0047] In another example, the heat exchangers disclosed herein may include one or more features to facilitate the containment of fluids held therein, such as the containment of any leak of a radioactive material from said fluids. For example, the heat exchangers of the present disclosure may include an outer shell, and a first inner tube and a second inner tube each held within the outer shell. The first inner tube may be configured to carry a first fluid (e.g., a molten fuel salt) and the second inner tube may be configured to carry a second fluid (e.g., a coolant salt). The first inner tube and the second inner tube may be arranged relative to one another in order to facilitate the transfer of heat between, for example, the fuel salt of the first inner tube to the coolant salt of the second inner tube. The outer shell may be fitted around the first inner tube and the second inner tube, such as being fitting around a portion of the first inner tube and a corresponding portion of the second inner tube that is configured to transfer heat therebetween, and serve as a containment structure in the event of a leaks from either tube. Further, in such arrangement, neither the first nor the second inner tube may be bathed, externally in a corrosive fluid. Accordingly, the inner tubes, and heat exchanger more generally, may exhibit enhance resistance to corrosive materials, which may prolong the service life of the heat exchanger. In some cases, one or both of the first inner tube and/or the second inner tube may be selectively removable from the outer tube, as described herein. In this regard, the respective inner tubes may be replaceable components of the heat exchanger, and may therefore allow the heat exchanger to be maintained and repaired during a shutdown event without necessarily requiring the entire heat exchanger to be removed from the molten salt system.

[0048] Turning to the Drawings, FIG. 1 depicts an example molten salt reactor system **100**. The molten salt reactor system **100** is depicted and described herein to illustrate example process equipment with which the various heat exchangers and associated apparatuses and systems of the present disclosure may be used. Accordingly, while the molten salt reactor system **100** is described herein, it will be appreciated that such heat exchangers and associated apparatuses and systems may be used with a variety of process equipment to detect leaks of conducting fluid being carried therethrough, as described herein.

[0049] With reference to the molten salt reactor system **100** of FIG. 1, the example molten salt reactor system **100** of FIG. 1 utilizes fuel salt enriched with uranium (e.g., high-assay low-enriched uranium) to create thermal power via nuclear fission reactions. In at least one example, the composition of the fuel salt may be $\text{LiF—BeF}_{2.2}\text{—UF}_{4.4}$, though other compositions of fuel salts may be utilized as fuel salts within the reactor system **100**. The fuel salt within the system **100** is heated to high temperatures (such as 600°C . or greater) and melts as the system **100** is heated.

[0050] As shown in FIG. 1, the molten salt reactor system **100** includes a reactor vessel **104** where the nuclear reactions occur within the molten fuel salt, a fuel salt pump **106** that pumps the molten fuel salt to a heat exchanger **110**, such that the molten fuel salt re-enters the reactor vessel after flowing through the heat exchanger **110**, and piping in between each component (e.g., piping **112a**, **112b**, **112c**, **112d**, **112e**). The molten salt reactor system **100** may also include additional components, such as, but not limited to, drain tank **108** and reactor access vessel **102**. The drain tank **108** may be configured to store the fuel salt once the fuel salt is in the reactor system **100** but in a subcritical state, and also acts as storage for the fuel salt if power is lost in the system **100**. The reactor access vessel **102** may be configured to allow for introduction of small pellets of uranium fluoride ($\text{UF}_{4.4}$) to the system **100** as necessary to bring the reactor to a critical state and compensate for depletion of fissile material. In several examples, the molten salt reactor system

100 may include an inert gas system and/or an equalization system (not shown in FIG. 1) to provide inert gas to a head space of the various salt-bearing components of the system **100** and to equalize pressures therebetween as needed for a given operation of the system **100**.

[0051] FIG. 1 further shows the system **100** as including an internal vessel or shield **120** that defines a first thermally insulative region **124** about select components of the system **100**. FIG. 1 further shows the system **100** as including a reactor enclosure **130**. The reactor enclosure **130** may be constructed from a thermally insulative metal (including certain stainless steels) that is capable of withstanding substantially high temperatures, such as temperature in excess of 600° C. The reactor enclosure **130** is shown, schematically, as encompassing the entirety of the internal shield **120** and any other salt-bearing components that are not otherwise included with the internal shield **120**. For example, the reactor enclosure **130** may define a second thermally insulative region **134** that receives the internal shield **120** and all the salt-bearing components that are not held within the first thermally insulative region **124**. The internal shield **120** and the reactor enclosure **130** may therefore each define a containment barrier about the salt-bearing components of the system **100**. Further, the internal shield **120** and the reactor enclosure **130** may define a substantially high-radiation and high-temperature zone of the system **100**.

[0052] The system **100**, as described herein, may be configured to generate heat energy via nuclear fission reactions with the molten fuel salt that is circulated through, among other components, the reactor vessel **104**. For example, the molten fuel salt may circulate through a fuel salt system **101** or “loop” including the reactor vessel **104**, the primary access vessel **102**, the pump **106**, and the heat exchanger **110**. Broadly, the molten fuel salt may exhibit an elevated temperature at an exit of the reactor vessel **104** as a result of fission reactions occurring therein. Such elevated temperature molten fuel salt may circulate through the fuel salt system **101** until the salt reaches the heat exchanger **110**. At the heat exchanger **110**, said heat may be extracted from the fuel salt. In turn, the lower temperature molten fuel salt may continue to circulate through the fuel salt system **101** until it reaches the entrance to the reactor vessel **104**, within which the salt may again increase in temperature via nuclear fission reactions.

[0053] The heat may be extracted from the fuel salt at the heat exchanger **110**, in part, using a coolant salt system **160**. The coolant salt system **160** may circulate a coolant salt through the heat exchanger **110** (also referred to herein as a “primary” heat exchanger). For example, and as shown in FIG. 1, the coolant salt system **160** may include a secondary heat exchanger **162** and a coolant pump **172**. The secondary heat exchanger **162** may be any appropriate heat exchanger, such as a shell and tube type heat exchanger, and be configured to circulate a coolant salt with the primary heat exchanger **110** via a flow induced by the coolant pump **172**. For example, and as shown in FIG. 1, the coolant salt system **160** may include a primary coolant loop cold legs **166a**, **166b** whereby a lower-temperature coolant is induced toward and into the primary heat exchanger **110** via the coolant pump **172**. The lower temperature coolant may receive heat from the high temperature fuel salt within the primary heat exchanger **110**. In turn, the coolant salt may exit the primary heat exchanger **110** having an elevated temperature, and may proceed to the secondary heat exchanger **162** via the secondary coolant loop hot leg **168**. The secondary heat exchanger **162** may, in turn, operate to extract heat from the primary coolant, and transfer said heat to another medium, such that the heat may be used to drive one or more processes (e.g., electrical power generation, chemical processing, and so on). For example, and as shown in FIG. 1, the secondary heat exchanger **162** may be fluidically coupled with a secondary coolant loop cold leg **170** whereat a lower temperature coolant medium may enter the secondary heat exchanger **162**. As further shown in FIG. 1, the secondary heat exchanger **162** may be fluidically coupled with a secondary coolant loop hot leg **168** whereat a higher temperature coolant medium may exit the secondary heat exchanger **162** after receiving heat from the primary coolant within the secondary heat exchanger **162**.

[0054] The primary heat exchanger **110**, the secondary heat exchanger **162** and/or any other heat

exchangers included with or associated with the molten salt system **100** may be configured as a heat exchanger of the present disclosure. For example, the primary heat exchanger **110**, the secondary heat exchanger **162** and/or other heat exchangers may be configured to permit gravitational draining of one or more fluids of salt therein, including the gravitational draining of at least one of a molten fuel salt or coolant salt held therein. In this regard, the primary heat exchanger **110**, the secondary heat exchanger **162**, and/or any other heat exchanger may be configured in a manner to support the exit of potentially radioactive materials therefrom upon a shutdown event, and thereby support the maintenance, repair and replacement of said heat exchangers and other components of the molten salt system. In this regard, it will be appreciated that the primary heat exchanger **110**, the second heat exchanger **162** and/or any other heat exchangers of the molten salt system may take the form of the various heat exchangers described in turn below in relation to FIGS. 2A-5C and/or may take the form of other variations of heat exchangers whereby said heat exchanger is configured to permit gravitational draining upon a shutdown event. In this regard, while examples of such heat exchangers are described in turn below in relation to FIGS. 2A-5C, it will be appreciated that in other examples, other arrangements and configured of heat exchangers that permit gravitational draining of molten fuel salt and/or coolant salt are possible and contemplated herein.

[0055] With reference to FIGS. 2A and 2B, an example heat exchange **200** is shown. The example heat exchanger **200** may be used in the molten salt system **100** as the primary heat exchanger **110**, the secondary heat exchanger **162**, and/or other heat exchanger of the system **100**. The heat exchanger **200** is shown in FIG. 2B in cross-section, taken along line 2B-2B of FIG. 2A. The heat exchanger **200** is shown in FIGS. 2A and 2B relative to a horizontal axis **201**. The horizontal axis **201** may be perpendicular to a gravitational direction of the heat exchanger **200**. The heat exchanger **200** may therefore be configured to permit gravitational draining of fluids (e.g., molten salts therein) by arranging one or more inlets or outlets thereof offset or oblique from the horizontal axis **201**.

[0056] The heat exchanger **200** may generally be constructed as a shell and tube type heat exchanger having an outer shell **202** and an inner tube **242** passing therein. The outer shell **202** may be configured as a pressure vessel configured to withstand temperatures in excess of 600° C. and pressures sufficiently above atmospheric. The outer shell **202** may be constructed from certain metal materials, including stainless steel and/or other corrosivity resistant materials. The outer shell **202** is shown in FIG. 2B as including a cylindrical region **204a**, and end cap regions **204b**, **204c**. The end cap regions **204b**, **204c** may each be connected (e.g., welded) to opposing ends of the cylindrical region **204a** in order to form the shell volume **220** therein. The shell volume **220** may be configured to circulate a fluid therethrough for heat transfer to a fluid of the inner tube **242**. In this regard, and as shown in further detail in FIG. 2B, the outer shell **202** may include a shell inlet **206** having an inlet passage **207** therethrough and a shell outlet **208** having an outlet passage **209** therethrough. The inlet passage **207** may extend through the shell inlet **206** to provide a fluid passage between a portion of the shell volume **220** and an environment external thereto (such as associated process piping). Further, the outlet passage **209** may extend through the shell outlet **208** to provide a fluid passage between a portion of the shell volume **220** and an environment external thereto (such as associated process piping). The outer shell **202** is further shown in FIG. 2B as including a baffle structure **230** held therein. The baffle structure **230** may include one or more elongated structures extending from an internal wall of the outer shell **202** in order to define a flow path through the shell volume **220**. For example, the baffle structure **230** may include a partition wall or other obstructing feature that extends from a baffle first end **232** attached to said inner wall, to a baffle second end **234** that may be a free end arranged within the shell volume **220**. In some cases, as described herein, the baffle second end **234** may be used to structurally support or otherwise engage with a portion of the inner tube **242**. The arrangement of the baffle **230** shown in FIG. 2B defines a flow or circulation path through the shell volume **220** between the shell inlet **206**

and the shell outlet **208**. For example, the baffle **230** may define a hot region **222** proximal the shell inlet **206** (where the outer shell **220** is used to receive elevated temperature fluid), a heat transfer region **224** proximal the baffle end **234**, and a cold region **226** proximal the shell outlet **208**. In this regard, fluid may flow along the circulation path from the hot region **222**, to the heat transfer region **224**, and to the cold region **226** with the shell volume **220** of the outer shell **202**. It will be appreciated that where the outer shell **220** is used to receive lower temperature fluid the region **222** may be “hot” and the region **226** may be “cold”.

[0057] To facilitate the temperature change of the fluid between the hot region **222** and the cold region **226**, the inner tube **242** may be arranged substantially within the heat transfer region **224** of the shell volume **220**. The inner tube **242** is shown in FIG. 2B as proceeding through a thickness of a wall of the outer shell **2202** and extending through the shell volume **220** along a U-shaped path generally oblique to the horizontal axis **201**. In other cases, the inner tube **242** may have a plurality of bends within the shell volume **220**. The inner tube **242** may therefore define a tube inlet **244** having an inlet passage **245** extending therethrough and a tube outlet **246** having an outlet passage **247** extending therethrough. The tube inlet **244** may be a structural portion of the inner tube **242** that extends through a wall of the outer shell **202** such that the inlet passage **245** fluidically couples a tube volume **250** with an environment external thereto (such as associated process piping).

Further, the tube outlet **246** may be a structural portion of the outer tube **242** that extends through a wall of the outer shell **202** such that outlet passage **247** fluidically couples the tube volume **250** with an environment external thereto (such as associated process piping). In this regard, fluid (e.g., a molten salt) may be circulated through the tube volume and be exposed to heat with the heat transfer region **224** of the shell volume **220**. For example, fluid contained in the heat transfer region **224** may bathe all or substantially all of the inner tube **242** that is held within the heat transfer region **224**, such as bathing first and second surfaces **243a**, **243b** of the inner tube **242**. The bathing of such surfaces may permit the transfer of heat from the hotter of the fluids to the cooler of the fluids substantially within the heat transfer region **224**.

[0058] The heat exchanger **200** may be coupled within the molten salt system (e.g., such as the system **200**) via various flanged connections, such as those shown in FIG. 2A. For example, the shell inlet **206** and the shell outlet **208** may include flanged connections **210a**, **210b**, respectively. The flanged connections **210a**, **210b** may be configured for fluidic coupling with corresponding flanged connections **292a**, **292b** of fuel salt piping **290a**, **290b** and/or other service piping based on a configuration of the heat exchanger **200** within the molten salt system. Additionally, as shown FIG. 2A, the tube inlet **244** and the tube outlet **246** may include flanged connections **248a**, **248b** respectively. The flanged connections **248a**, **248b** may be configured for fluidic coupling with corresponding flanged connections **298a**, **298b** of coolant salt piping **296a**, **296b** and/or other service piping based on a configuration of the heat exchanger **200** within the molten salt system **200**.

[0059] As shown in FIGS. 2A and 2B, the heat exchanger **200** may be configured to permit gravitational draining of at least one of a molten fuel salt or a coolant salt upon a shutdown event. For example, each of shell inlet **206** and shell outlet **208** are shown arranged oblique to the horizontal axis **201**. In this regard, fluid held within the shell volume **220** may be encouraged to exit the shell volume **220** via gravitational flow from the shell volume **220** and through the inlet passage **207** and outlet passage **209**. Further, each of the tube inlet **244** and the tube outlet **246** are shown arranged oblique to the horizontal axis **201**. In this regard, fluid held within the tube volume **250** may be encouraged to exit the tube volume **250** via gravitational flow from the tube volume **250** and through the inlet passage **245** and outlet passage **248**. Accordingly, to the extent that either shell volume **220** or the tube volume **250** includes molten materials, rather than solidify therein upon a shutdown event, said molten materials may be encouraged to gravitationally exit said volumes. Because the molten materials may exit the respective volumes, the heat exchanger **200** may support the maintenance, repair, and replacement thereof because the heat exchanger **200** may

have a reduce amount of solidified radioactive material held therein.

[0060] With reference to FIG. 3A and 3B, an example heat exchanger **300** is shown. The example heat exchanger **300** may be used in the molten salt system **100** as the primary heat exchanger **110**, the second heat exchanger **162**, and/or other heat exchanger of the system **100**. The heat exchanger **300** is shown in FIG. 3B in cross-section, taken along line 3B-3B of FIG. 3A. The heat exchanger **300** is shown in FIGS. 3A and 3B relative to a horizontal axis **301**. The horizontal axis **301** may be perpendicular to a gravitational direction of the heat exchanger **300**. The heat exchanger **300** may therefore be configured to permit gravitation draining of fluids (e.g., molten salts therein) by arranging one or more inlets or outlets thereof offset or oblique from the horizontal axis **301**. It will be appreciated that in some cases, the one or more inlets or outlets thereof may be arranged perpendicular from the horizontal axis **301** such that the respective inlets and outlets are generally arranged along a vertical axis of the heat exchanger **300** to support gravitational draining therealong. In this regard, the heat exchanger **300** may be substantially analogous to the heat exchanger **200** may include: an outer shell **302**, a cylindrical region **304a**, end cap regions **304b**, **304c**, a shell inlet **206**, an inlet passage **207**, flanged connections **310a**, **310b**, a shell volume **320**, a cold region **322**, a heat transfer region **324**, a hot region **326**, a baffle structure **330**, a first baffle end **332**, a second baffle end **334**, an inner tube **342**, a first surface **343a**, a second surface **343b**, a tube inlet **344**, an inlet passage **345**, a tube outlet **346**, an outlet passage **347**, flanged connections **348a**, **348b**, a tube volume **350**, a heat transfer region **352**, fuel salt piping **390a**, **390b**, flanged connections **392a**, **392b**, coolant salt piping **396a**, **396b**, and flanged connections **398a**, **398b**; redundant explanation of which is omitted here for clarity.

[0061] Notwithstanding the foregoing similarities, the shell inlet **306** and the shell outlet **308** are shown arranged along a common direction as the tube inlet **344** and the tube outlet **346**. In this regard, both the shell volume **320** and the tube volume **350** may be configured to gravitationally drain along the same direction. In FIGS. 3A and 3B, the respective inlets and outlets are shown oblique to the horizontal axis **301**; in other cases, such inlets and outlets may be arranged perpendicular to the horizontal axis **301** such that the inlets and outlets are each substantially aligned with a vertical axis of the heat exchanger **300**. In turn, the heat exchanger **300** may facilitate maintenance, repair and replacement thereof as said fluids (e.g., molten salts) drain along a common direction, and as such, the heat exchanger **300** may be more readily uncoupled from the molten salt system. To facilitate the arrangement of the shell inlet **306**, the shell outlet **308**, the tube inlet **344** and the tube outlet **346** as being arranged along a common direction, the inner tube **342** may extend through the shell volume **320** along a U-shaped path about and over the baffle structure **330**, as shown in FIG. 3B. Such configuration may allow the fluid held within the shell volume **320** to bathe a greater surface area of the inner tube **342**, and thereby promote the efficient heat transfer between the two fluids of the respective volumes.

[0062] With reference to FIGS. 4A and 4B, an example heat exchanger **400** is shown. The example heat exchanger **400** may be used in the molten salt system **100** as the primary heat exchanger **110**, the secondary heat exchanger **162**, and/or other heat exchanger of the system **100**. The heat exchanger **400** is shown in FIG. 4B in a cutaway view of the heat exchanger of FIG. 4A. The heat exchanger **400** is shown in FIGS. 4A and 4B relative to a horizontal axis **401**. The horizontal axis **401** may be perpendicular to a gravitational direction of the heat exchanger **400**. The heat exchanger **400** may therefore be configured to permit gravitation draining of fluids (e.g., molten salts therein) by arranging one or more inlets or outlets thereof offset or oblique from the horizontal axis **401**. In this regard, the heat exchanger **400** may be substantially analogous to the heat exchanger **200**, and may include: an outer shell **402**, a cylindrical region **404a**, end cap regions **404b**, **404c**, flanged connections **410a**, **410b**, an inner tube **442**, a first surface **443a**, a second surface **443b**, a tube inlet **444**, an inlet passage **445**, a tube outlet **446**, an outlet passage **447**, flanged connections **448a**, **448b**, tube volume **450**, a heat transfer region **452**, fuel salt piping **490a**, **490b**, flanged connections **492a**, **492b**, coolant salt piping **496a**, **496b**, and flanged connections

498a, 498b; redundant explanation of which is omitted here for clarity.

[0063] Notwithstanding the foregoing similarities, the heat exchanger **400** is arranged with a shell tube **420** to carry the molten fuel salt (or other fluid) through the outer shell **402**. In this regard, the outer shell **402** may operate as a containment structure defining a containment volume **419** within which a portion of the shell tube **420** and the inner tube **442** are arranged. For example, the shell tube **420** may define a U-shaped tube extending through the containment volume between a shell tube inlet **406** and a shell tube outlet **408**. In the regard, the shell tube **420** may have a volume **423** which extends between an inlet passage **407** and an outlet passage **409** and which is fluidically separated from the containment volume **419**. In this regard, fluid may progress through the volume **423** and not within the containment volume **419** so as to not bathe the inner tube **442**.

Notwithstanding, the shell tube **420** may generally define a hot region **422**, a heat transfer region **424**, and a cold region **426**. Fluid (such as a heated molten salt) may therefore proceed into the heat exchanger **400** via the inlet passage **407** and along the hot region **422**. At the heat transfer region **424**, the inner tube **442** (which may carry a relatively cooler coolant salt) may be arranged such that the heat may transfer from the fluid of the shell tube **420** to the fluid of inner tube **442**. In some cases, surfaces **421a**, **421b** of the shell tube **420** maybe in contact with surfaces **443a**, **443b** to facilitate the foregoing. The fluid of the shell tube **420** may be subsequently exit the volume **423** via the outlet passage **409** of the shell tube outlet **408**.

[0064] With reference to FIGS. 5A-5C, an example heat exchanger **500** is shown. The example heat exchanger **500** may be used in the molten salt system **100** as the primary heat exchanger **110**, the secondary heat exchanger **162**, and/or other heat exchanger of the system **100**. The heat exchanger **500** is shown in FIG. 5B in cross-section, taken along line 5B-5B of FIG. 5A. The heat exchanger **500** is shown in FIG. 5C in an exploded view configuration showing select components removed from one another, as described herein. The heat exchanger **500** is shown relative to a horizontal axis **501**. The horizontal axis **501** may be perpendicular to a gravitation direction of the heat exchanger **500**. The heat exchanger **500** may therefore be configured to permit gravitational draining of fluid (e.g., molten salts therein) by arranging one or more inlet or outlets thereof offset or oblique from the horizontal axis **501**. In this regard, the heat exchanger **500** may be substantially analogous to the heat exchanger **400**, and may include: an outer shell **502**, a cylindrical region **504a**, end cap regions **504b**, **504c**, a shell inlet **506**, an inlet passage **507**, a shell outlet **508**, an outlet passage **509**, flanged connections **510a**, **510b**, a containment volume **519**, a shell tube **520**, a first surface **521a**, a second surface **521b**, a cold region **522**, a volume **523**, a heat transfer region **524**, a hot region **526**, an inner tube **542**, a first surface **543a**, a second surface **543b**, a tube inlet **544**, an inlet passage **545**, a tube outlet **546**, an outlet passage **547**, flanged connections **548a**, **548b**, a tube volume **550**, heat transfer region **552**, fuel salt piping **590a**, **590b**, flanged connections **592a**, **592b**, coolant salt piping **596a**, **596b**, and flanged connections **598a**, **598b**; redundant explanation of which is omitted here for clarity.

[0065] Notwithstanding the foregoing similarities, the heat exchanger **500** may be configured to removably replace the inner tube **542**. For example, the inner tube **542** may be a removably replaceable component of the heat exchanger **500** in order to support the maintenance, repair, and replacement of the heat exchanger **500** and tubes therein. In this regard, as shown in FIG. 5C, the inner tube **542** may be coupled with a cover **584**. The cover **584** may be a flange or other structural feature through which the inner tube **542** may pass through and enter the containment volume **523**. For example, the tube inlet **544** may pass through the cover **584** at an interface **586a**. Further, the tube outlet **546** may pass through the cover **584** at an interface **586b**. The inner tube **542** may be selectively positioned within the heat exchanger **500** by engagement of the cover **584** with a transition piece **580**. The transition piece **580** may be an oblique pipe, fitting, and/or other component having an interface passage **582** that extends from an exterior of the heat exchanger **500** and into the containment volume **519**. In operation, the inner tube **542** may be positioned at least partially within the containment volume **519** via the interface passage **582**. In turn, the cover **584**

may be removably secured to a face **581** of the transition piece **580**, such as via bolts and/or other fasteners, in order to seal the inner tube **542** within the containment volume **519**. In this regard, upon a shutdown event, the foregoing process may be reversed in order to remove the inner tube **542** from the containment volume **519** within necessarily disturbing the remainder of the heat exchanger **500**. For example, the cover **584** may be uncoupled from the face **581**, and the inner tube **542** removed from the containment volume **519** while the heat exchanger **500** remains in place with the molten salt system (e.g., the system **100**).

[0066] With reference to FIGS. **6A-6B**, and example heat exchanger **612** installed into a molten salt reactor system **600** is shown. Molten salt reactor system **600** may be substantially similar to molten salt reactor system **100** of FIG. **1** and include a reactor vessel **604**, a reactor access vessel **602**, a primary heat exchanger **622**, piping therebetween (**606a**, **606b**), and a reactor enclosure **608** redundant explanation of which is excluded herein for clarity. Notwithstanding the foregoing similarities, reactor system **600** may include a double-layered reactor enclosure **608** including a thermal insulation layer **610** and a radiation shielding layer **612** and a gap **614** interposed therebetween.

[0067] Heat exchanger **622** may be used as the primary heat exchanger **110** of molten salt reactor system **100**. The heat exchanger **622** is shown relative to horizontal axis **620**. The horizontal axis **620** may be perpendicular to a gravitation direction of the heat exchanger **622**. The heat exchanger **620** may therefore be configured to permit gravitational draining of fluid (e.g., molten salts therein) by arranging one or more inlet or outlets thereof offset or oblique from the horizontal axis **620**. In this regard, the heat exchanger **622** may be substantially analogous to the aforementioned heat exchangers and include an outer shell **624**, a shell inlet **626**, an inlet passage **628**, a shell volume **630**, a cold region **632**, a heat transfer region **634**, a hot region **636**, inner tubes **638**, tube inlets **640**, tube outlets **642**, and coolant salt piping **644a**, **644b** redundant explanation of which is omitted here for clarity.

[0068] However, FIGS. **6A-6B** highlight other components that may be included to install heat exchanger **622** into a reactor system. For example, heat exchanger **622** may further include a waterbox **646** configured to enable passage of a coolant into the plurality of tubes **638**, a tube sheet **648** to hold the plurality of tubes **638**, a partition **650** within waterbox **646** to separate incoming coolant from outcoming coolant.

[0069] Additionally, heat exchanger **622** may be mounted within the molten salt reactor system **600** and to the reactor enclosure **608** via a flange assembly **652**. Flange assembly **652** may be operable to mount heat exchanger **622** into reactor system **600** such that it may facilitate heat transfer functions. Flange assembly **652** is further operable to facilitate removal of heat exchanger **622** for maintenance purposes, following gravitational draining as described herein. In this regard, flange assembly **652** may include a first flange **654** and a second flange **656**. First flange **654** may be attached to thermal insulation layer **610** via bolt connections **660a** while second flange **656** may be attached to radiation shielding layer **612** via bolt connections **600b**. Interposed between first flange **654** and second flange **656** and facilitating a connection therebetween is penetrator **658**. Penetrator **658** may generally extend from second flange **656**, through gap **614**, and to first flange **654**. In this regard, flange assembly **652** enables access to first flange **654**, such that it may be unbolted from thermal insulation layer **610** enabling removal of heat exchanger **622**. Additionally, penetrator **658** enables such access while maintaining a continuous seal of reactor enclosure **608**. In one example, heat exchanger **622** may be disconnected from piping **606a**, **606b** following gravitational draining. Then, flange **654** may be disconnected from thermal insulation layer **610** enabling an operator to pull heat exchanger **622** from reactor system **600**. In another example, following gravitational draining and piping disconnection, flange **654** and flange **656** may be disconnected such that the entire flange assembly **652** may be removed along with heat exchanger **622**.

[0070] FIG. **7** depicts a flow diagram of a method **700** of draining a heat exchanger of a molten salt reactor system. At operation **704**, a fuel salt system is operated. For example, and with reference to

FIG. 1, the fuel salt system **101** of the molten salt system **100** is operated. As described herein, a molten fuel salt may undergo nuclear fission reactions within a reactor vessel **104** that causes the molten fuel salt to exhibit an elevated temperature. The elevated temperature molten fuel salt may circulate through the fuel salt system **101** to the heat exchanger **110** whereat heat may be removed from the molten fuel. The molten fuel salt may continue circulation through the fuel salt system **101** and into the reactor vessel **104** whereat the molten fuel salt may again be heated through nuclear fission reactions.

[0071] At operation **708**, a coolant salt system is operated. For example, and with reference to FIG. 1, the coolant salt system **160** of the molten salt system **100** is operated. As described herein, a coolant salt (and/or other coolant medium) may be circulated through the heat exchanger **110** in order to remove heat from the molten fuel salt. For example, a coolant pump **172** may operate to induce a flow of a coolant to the heat exchanger **110** via coolant loop cold legs **166a**, **166b**. The coolant salt may receive heat from the heat exchanger **110**, and may continue along the coolant salt system **160** to a secondary heat exchanger **162** via a coolant loop hot leg **164**. The secondary heat exchanger **162** may operate to remove heat from the coolant salt for use in other processing, including electricity generation, chemical production and others. In this regard, FIG. 6 further shows an operation **712**, in which heat is transfer from the fuel salt system **101** to the coolant salt system **160** using the heat exchanger **110**.

[0072] At operation **716**, the molten fuel salt and the coolant salt is gravitationally drained from the heat exchanger upon a shutdown event. For example, and with reference to FIGS. 1 and 2B, a fuel salt and/or a coolant salt may be gravitationally drained from the heat exchanger **200**. For example, upon a shutdown event, fluid contained within the shell volume **220** may be encouraged to drain therefrom via the passages **207**, **209**, which are each offset from a horizontal direction of the heat exchanger **200**. Further, upon a shutdown event, fluid contained within the tube volume **250** may be encouraged to drain therefrom via the passages **245**, **247**, which are each offset from a horizontal direction of the heat exchanger **200**. Analogously, a fuel salt and/or a coolant salt may be gravitationally drained from any of the heat exchangers **300**, **400**, **500**, **622**. For example, upon a shutdown event, fluid contained within the respective volumes of the heat exchangers **300**, **400**, **500**, **622** may be encouraged to drain therefrom via the respective passages, which are offset from a horizontal direction of the corresponding heat exchangers. By permitting the gravitational draining of the fluids (molten salts) from the heat exchangers, said molten salt may be inhibited from solidifying therein, and thus the heat exchangers may have a reduced amount of radioactive material therein upon such shutdown event. The reduced amount of radioactive material within the heat exchanger upon the shutdown event may support the maintenance, repair, and/or replacement of the heat exchanger and select components. Following gravitational draining the heat exchangers may be removed from the reactor system via, for example, flange assembly **652**. Thereafter, the heat exchanger may undergo maintenance, repair and exchanged or may be simply replaced by a substantially similar heat exchanger.

[0073] Other examples and implementations are within the scope and spirit of the disclosure and appended claims. For example, features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations. The foregoing description, for purposes of explanation, uses specific nomenclature to provide a thorough understanding of the described examples. However, it will be apparent to one skilled in the art that the specific details are not required in order to practice the described examples. Thus, the foregoing descriptions of the specific examples described herein are presented for purposes of illustration and description. They are not targeted to be exhaustive or to limit the examples to the precise forms disclosed. It will be apparent to one of ordinary skill in the art that many modifications and variations are possible in view of the above teachings.

Claims

1. A molten salt reactor system comprising a fuel salt system configured to circulate a molten fuel salt through a molten salt loop; a coolant salt system configured to circulate a coolant salt through a coolant loop; and a heat exchanger arranged along the molten salt loop and the coolant loop and configured to transfer heat from the molten fuel salt of the molten salt loop to the coolant salt of the coolant loop, wherein the heat exchanger is configured to permit gravitational draining of at least one of the molten fuel salt or the coolant salt upon a shutdown event.
2. The molten salt reactor system of claim 1, wherein the heat exchanger comprises an outer shell defining a shell volume configured to receive one of the molten fuel salt or the coolant salt, and an internal tube positioned at least partially within the shell volume and configured to receive the other of the molten fuel salt or the coolant salt.
3. The molten salt reactor system of claim 2, wherein the outer shell comprises a baffle defining a shell circulation path about a portion of the internal tube positioned in the shell volume, and the heat exchanger is arranged on the molten salt loop and the coolant loop with the circulation path positioned off-set from a horizontal direction of the molten salt reactor system.
4. The molten salt reactor system of claim 3, wherein the internal tube extends oblique to the outer shell within the shell volume.
5. The molten salt reactor system of claim 4, wherein the internal tube comprises a plurality of bends within the internal tube volume, and the oblique orientation of the internal tube relative to the outer shell permits draining of the other of the molten fuel salt or the coolant salt upon the shutdown event.
6. The molten salt reactor system of claim 2, wherein the internal tube extends substantially along the shell circulation path.
7. The molten salt reactor system of claim 1, wherein the shell circulation path proceeds along a U-shaped path through the shell volume, the internal tube comprises a U-shaped tube extending along the U-shaped path, and an inlet and an outlet for the shell circulation path and an inlet and outlet for the internal tube are each arranged along a bottom side of the outer shell such that the respective inlets and outlets are each substantially aligned with a vertical direction of the molten salt reactor system.
8. The molten salt reactor system of claim 1, wherein the shutdown event is triggered based on one or more of a loss of power, a high radiation alarm and/or a containment breach.
9. The molten salt reactor system of claim 1, wherein the molten salt reactor system further comprises a reactor vessel, a pump, and a drain tank all arranged along the molten salt loop, and the reactor vessel outputs an elevated temperature form of the molten fuel salt to the heat exchanger.
10. The molten salt reactor system of claim 1, wherein the heat exchanger comprises a pair of oblique tubes associated with one another inside of an outer shell, a first oblique tube of the pair of oblique tubes is configured to receive one of the molten fuel salt or the coolant salt, and a second oblique tube of the pair of oblique tubes is configured to receive the other of the molten fuel salt or the coolant salt.
11. The molten salt reactor system of claim 10, wherein the first oblique tube is removably couplable from the outer shell and separatable from the first oblique tube.
12. The molten salt reactor system of claim 1, further comprising a flange assembly operable to facilitate removal of the heat exchanger from the molten salt reactor system.
13. A molten salt reactor system comprising a fuel salt system configured to circulate a molten fuel salt through a molten salt loop; a coolant salt system configured to circulate a coolant salt through a coolant loop; and a heat exchanger arranged along the molten salt loop and the coolant loop and configured to transfer heat from the molten fuel salt of the molten salt loop to the coolant salt of the coolant loop, wherein the heat exchanger comprises at least one oblique tube configured to route at

least one of the molten fuel salt or the coolant salt out of the heat exchanger upon a shutdown event.

14. The molten salt reactor system of claim 13, wherein the heat exchanger comprises a shell and tube heat exchanger, and the at least one oblique tube is arranged oblique to an outer shell of the shell and tube heat exchanger.

15. The molten salt reactor system of claim 13, wherein the molten fuel salt and the coolant salt begin to solidify at a solidification temperature that is below 550° C.

16. The molten salt reactor system of claim 13, wherein the at least one oblique tube comprises a first oblique tube configured to route the molten fuel salt out of the heat exchanger, and a second oblique tube configured to route the coolant salt out of the heat exchanger.

17. The molten salt reactor system of claim 16, wherein a portion of the first oblique tube and a corresponding portion of the second oblique tube are removably interposed with one another with an outer shell of the heat exchanger.

18. The molten salt reactor system of claim 13, further comprising a flange assembly operable to facilitate removal of the heat exchanger from the molten salt reactor system.

19. A method of draining a heat exchanger of a molten salt reactor system, the method comprising operating a fuel salt system by circulating a molten fuel salt through a molten salt loop; operating a coolant salt system by circulating a coolant salt through a coolant salt loop; transferring heat, using a heat exchanger, from the molten fuel salt of the molten salt loop to the coolant salt of the coolant salt loop; and gravitationally draining, upon a shutdown event, the molten fuel salt and the coolant salt from the heat exchanger.

20. The method of claim 19, wherein the heat exchanger comprises at least one oblique tube configured to route at least one of the molten fuel salt or the coolant salt through the heat exchanger during the transferring.

21. The method of claim 20, wherein said at least oblique tube is arranged oblique to an outer shell of the heat exchanger.

22. The method of claim 19, wherein the molten fuel salt includes a fissile material.
